

SAMMY ABDELLA

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New York City, NY

COMPUTER SCIENTIST

I am a results-driven software engineer with a passion for backend systems, data optimization, and scalable solutions. My background spans financial institutions, postal logistics, and academic research, with a consistent record of improving system efficiency, ensuring compliance, and enabling better decision-making through data.

Areas of Expertise

Backend Development • API Optimization • Microservices Architecture • Data Analysis • PostgreSQL Optimization • OAuth2/JWT • Docker & Kubernetes • Scientific Computing • Medical Software Tools • Research & Technical Writing • Multilingual Communication (English, French, Arabic)

EDUCATION

Bachelor of Science in Computer Science | Columbia University – New York City, NY

May 2027

PROFESSIONAL WORK HISTORY

2025 Web Developer Intern | Columbia College Information Technology

May 2025 – Present

- Migrated 15 Columbia departmental websites from Drupal 7 to Drupal 10, writing migration scripts, refactoring custom modules, and handling regression testing, improving average page load times by ~35% and eliminating 100% of deprecated modules in production
- Integrated a Django + PostgreSQL + RESTful API backend with a new Columbia admissions website, scaling reliably to 50,000+ applicants per cycle, cutting form submission errors by ~40%, and reducing manual data correction work by ~50%

Research | Columbia University Artificial Intelligence Laboratory

January 2025 – September 2025

- Lead engineer for gAyl, an AI tool for LGBTQ+ suicide prevention. Designed six model-based safety benchmarks to generate treatment plan recommendations and achieved <15% false-negative rate on unsafe-response detection across 1,000+ high-risk prompts
- Trained LLM-based classifiers in PyTorch and Hugging Face Transformers to automate safety evaluation, reducing manual benchmarking time by ~95%
- Publication: <https://doi.org/10.7916/z6vc-d867>

Software Developer at Zitouna Bank | Tunis, Tunisia – Remote

January 2024 – May 2024

- Engineered and optimized RESTful APIs using Python and Django, cutting response times by 30% and boosting transaction throughput by 25%
- Partnered with database teams to manage and tune PostgreSQL performance for real-time, high-volume financial data
- Ensured backend infrastructure aligned with security standards and financial regulations, resulting in a 100% compliance rate during audits

Freelance Software Developer | Fiverr – Remote

November 2021 – December 2021

- Assisted 50+ clients in customizing web applications, developing mobile apps, integrating APIs, optimizing website performance, and implementing cybersecurity measures, \$10,000+ total earnings

TECHNICAL SKILLS

Python, C++, Java, JavaScript, SQL, HTML/CSS, Django, RESTful APIs, OAuth2, JWT, Docker, Kubernetes, PostgreSQL, API Integration, Web Optimization,